Lossless Astronomical Image Compressors

Algorithmic Information Theory

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Motivation

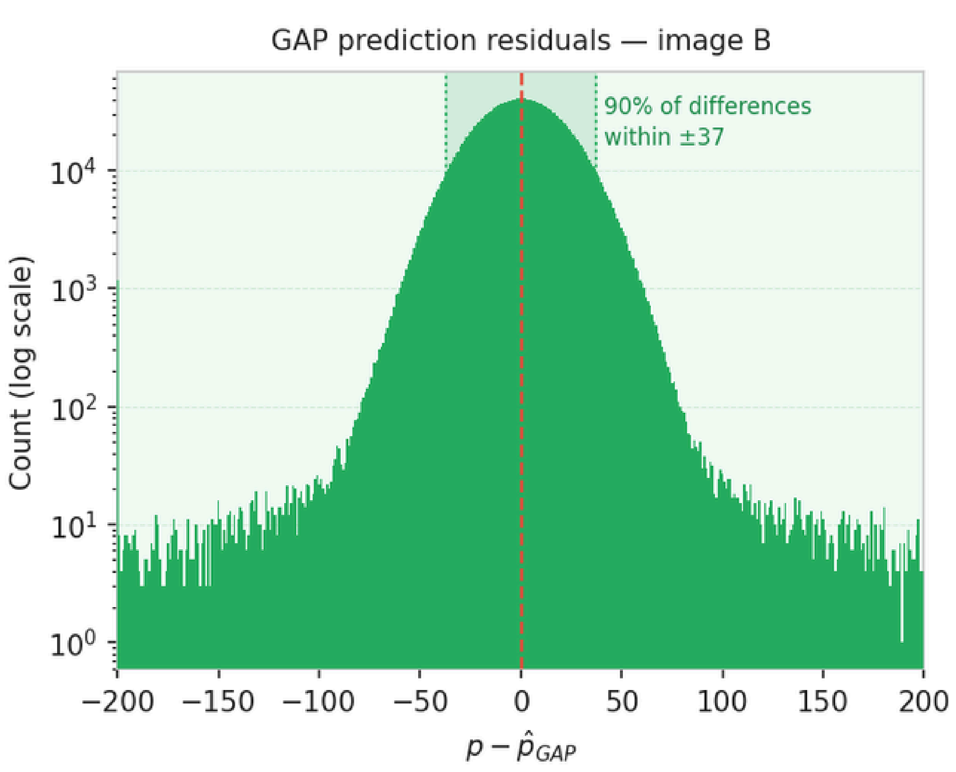
Lossless compression tools like gzip, xz, and zstd are designed to work well across any type of data. When the structure of the data is known in advance, however, a domain-specific compressor can exploit it directly, achieving better compression ratios and faster runtimes than any general-purpose alternative.

Astronomical images are a strong example of this: smooth backgrounds, sparse point sources, and high spatial correlation make them highly predictable. We design and evaluate three lossless codecs that exploit these properties, each targeting a different point on the speed-ratio trade-off.

Dataset

The benchmark corpus consists of 8 raw astronomical images (1500×1500, 16-bit big-endian, 4.5 MB each) from ground-based telescopes spanning bias frames, flat fields, and science frames. This diversity ensures no single predictor dominates results artificially.

Adjacent pixels in astronomical images are highly correlated, making spatial prediction a natural fit. Instead of storing raw pixel values, we predict each pixel from its neighbours and only encode the error. The plot shows that using GAP prediction most residuals are near zero, meaning the predictor almost always gets it right.



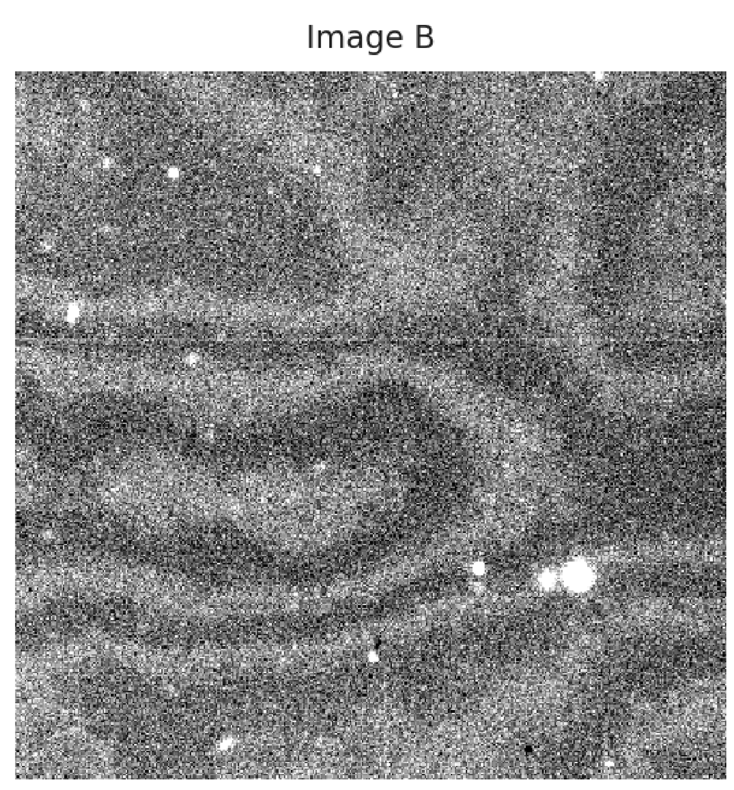


Fig. 1 - GAP prediction residuals for image B (1499×1499 pixels)

Fig. 2 - Crop of raw image B (400×400, 16-bit)

Design Strategy

All three codecs follow the same core pipeline. Each pixel is predicted from its already-decoded neighbours and only the error is stored. This residual is then split into a high byte and a low byte, which are coded independently. After a good prediction the high byte is almost always zero, so it compresses to almost nothing, leaving all meaningful variation in the low byte.

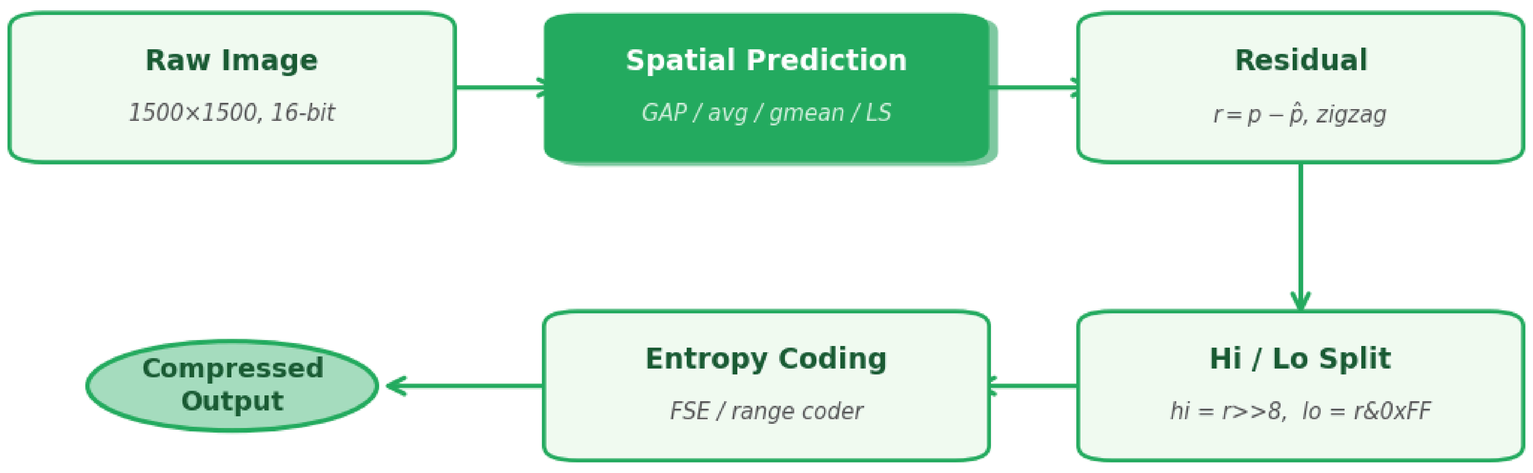


Fig. 3 - Common compression pipeline shared by all three codecs

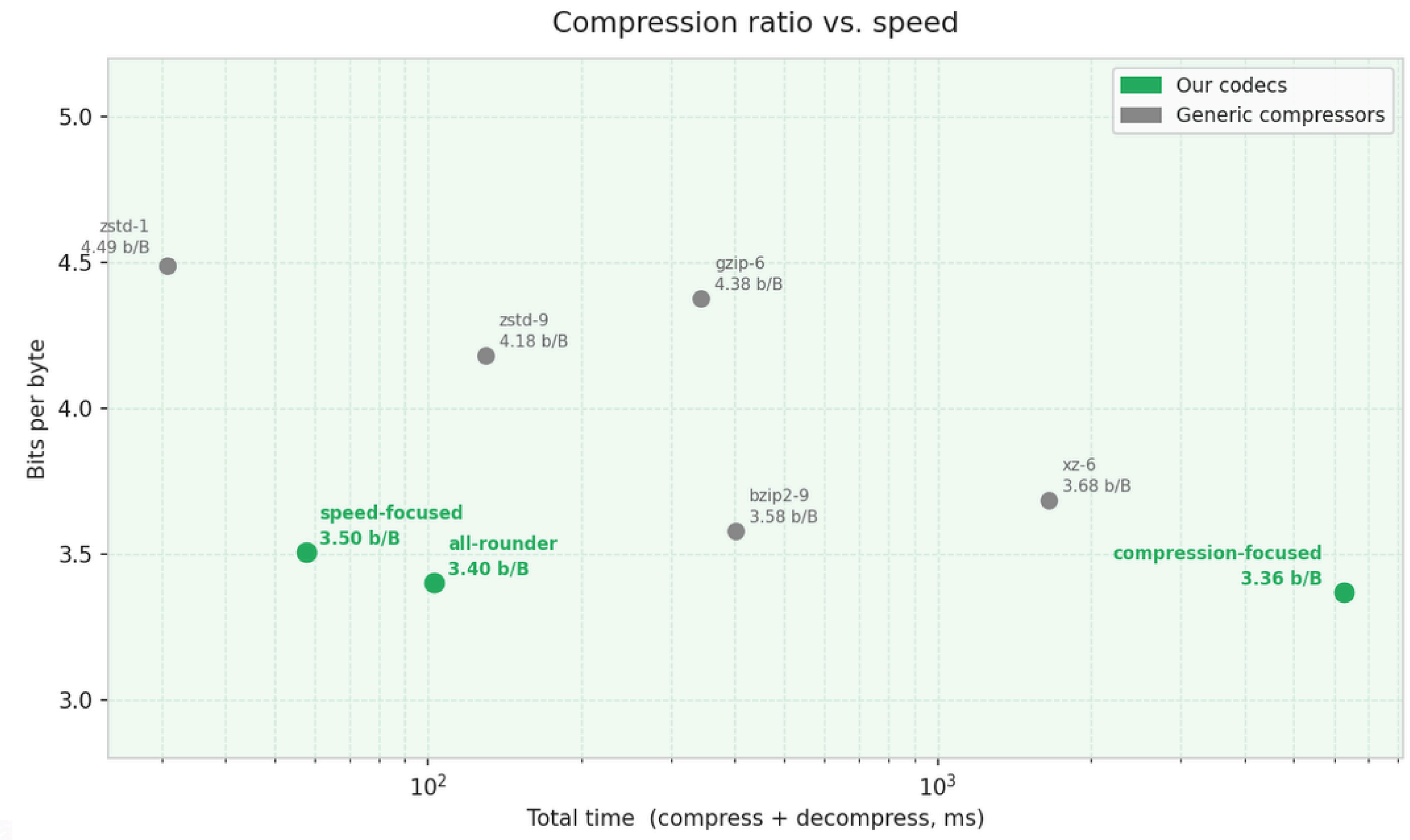
The three codecs differ in how aggressively they search for the best predictor. *Speed-Focused* applies GAP across parallel horizontal strips for maximum throughput. *All-Rounder* selects the best of five candidate predictors per block, including training-free least-squares. *Compression-Focused* exhausts 60 predictor and context combinations per block and tries seven block sizes, keeping the smallest result.

	Speed-Focused	All-Rounder	Compression-Focused
Predictor	GAP	Multi-mode adaptive (5 modes)	10 predictors (raw, avg, gmean, GAP, MED, LS-4 to LS-24)
Residual mapping	Zigzag	Zigzag	Zigzag
Entropy coder	Adaptive range	FSE/tANS	Adaptive range
Hi/Lo byte split	Yes	Yes	Yes
Lo-byte contexts	8	Context FSE	Up to 256 (auto-selected)
Block selection	Fixed strips	Per-block mode selection	7 block sizes
Parallelism	Row strips	Square blocks	Square blocks + 7 runs

Fig. 4 - Comparison of the three codec designs

Results and Conclusions

All three codecs beat every tested general-purpose compressor in compression ratio. The Speed-Focused codec achieves this in just 23 ms, the All-Rounder balances ratio and speed at 59 ms, and the Compression-Focused codec reaches the best ratio of 3.36 bpp.



Compressor	Avg bpp	Ratio	Avg compress (ms)	Avg decompress (ms)
Speed-Focused	3.50	43.7%	23	34
All-Rounder	3.40	42.5%	59	44
Compression-Focused	3.36	42.0%	6156	133

Fig. 5 - Comparison of lossless compressors on the benchmark dataset (8 images, 1500×1500, 16-bit)

All three codecs outperform every tested generic compressor in compression ratio while two of three also beat bzip2-9 in total runtime. The consistent margin across all eight benchmark images confirms that the gains are structural and not dataset-specific. This makes spatial prediction on native 16-bit values the natural approach for astronomical image compression.

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